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1D model for low power soliton hybridized with a plasmon in realistic nonlinear planar structures

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Abstract. We study the propagation of nonlinear waves in layered nonlinear dielectric/linear dielectric/metal planar structures. We generalyze a 1D vector model that describes the light propagation in such configurations embedded in a linear dielectric medium. We compute the nonlinear dispersion relation and the field profiles. We use our model to design realistic structures, in terms of linear and nonlinear properties, which support soliton waves with a plasmon tail at low peak power around or below 1GW/cm^2. These results open the way for potential observation of such states in chalcogenide waveguides associated with silica and metal films. A recordable plasmonic part of the field extends in the external medium.

Keywords: spatial soliton, plasmon, nonlinear wave, nonlinear dispersion relation, Kerr effect, planar structure, metal film
PACS: 03.75.Lm, 73.20.Mf, 82.35.Ej, 42.65.Hw, 42.65.Wi, 78.66.Fd

INTRODUCTION

It is now well established that plasmonics is more than an emerging research field. A lot of works have already been published [1]. Similarly, optical solitons have been studied since more than forty years and they are still an active research field [2]. Several works have been recently published about self-sustained solutions that combine soliton and plasmon features in metal/nonlinear dielectric structures [3, 4, 5] and are essentially different from nonlinear plasmonics in which it is metal that works on the nonlinear properties. Devices that support nonlinear plasmon-soliton waves may be of interest for sensor applications. The input power dependency can be considered as a new degree of freedom.

Actually, the first description of 1D nonlinear plasmon-soliton waves propagation along metal/nonlinear dielectric interfaces was given more that 30 years ago [6, 7, 8, 9] though a different terminology was used to describe such states (see. Fig. 2 in [6]). In these structures, the nonlinear stationary solution is made of a spatial soliton part coupled with a plasmonic wave. Lately, the interest in such plasmons-solitons started to grow again [3, 4, 5, 10, 11].

Nevertheless, up to now no experimental results have been published on the issue of plasmon-soliton coupling. The main reason is that, for the previously proposed structures, the nonlinear refractive index change required for the formation of plasmon-soliton waves is too high compared with the one attainable for real materials or, equivalently, the peak power is too high when ones uses realistic values of nonlinear coefficients of integrated optics conventional materials [2, 12].

In the present work, we describe for the first time planar structures made of conventional materials supporting a low peak power nonlinear solutions that combine a soliton profile with a measurable plasmonic field. In its current form, it is made of a chalcogenide glass coated with silica and gold films.

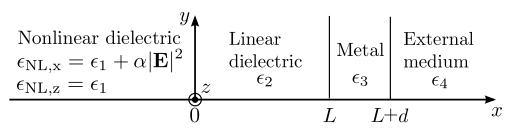

FIGURE 1. Geometry of the 4-layer nonlinear model used to study 3-layer structures. ε_j denotes the linear permittivity of j-th layer ($j \in \{1,2,3,4\}$).

It is worth mentioning that structures similar to the ones we propose have already been fabricated even if it was not in the framework of soliton-plasmon studies and that optical solitons have already been observed in chalcogenide planar waveguide for peak power around 2GW/cm^2 [13]. Furthermore, the configuration we propose can be made suitable for sensor applications since the decaying part of the plasmon field can be located at a metal/air or at a metal/water interface.

In our case, the structure is designed to make the plasmon field at the interface between the metal layer and the external dielectric medium recordable using near field optics measurements, and to allow its generation and control by beams with low peak intensities.

A vector approach is required to study such nonlinear waves that include a plasmon part. In our work we expand the 1D vector model used for 3-layer symmetric configurations made of a finite thickness metal layer embedded in semi-infinite nonlinear dielectric regions [6]

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to study the propagation of nonlinear waves in a 4-layer configuration (see Fig. 1). This 4-layer model is required so as to describe our 3-layer structure coupled with a semi-infinite external medium. We found out that to couple in the same wave a pronounced soliton with a plasmon that extends in a linear low-index external medium a nonlinear dielectric/linear dielectric/metal structure is needed.

1D NONLINEAR MODEL FOR 4 LAYERS

Let us consider the propagation of nonlinear light waves in a layered metal/nonlinear dielectric/linear dielectric structure presented in Fig. 1. We consider a Kerr type nonlinearity of the form $\alpha(x) = \mathscr{H}(-x)\varepsilon_0 c\varepsilon(x)n_2$, where $\mathscr{H}(x)$ denotes the Heaviside step function, ε_0 the vacuum permittivity, c the speed of light, n_2 is the Kerr nonlinear coefficient, and $\varepsilon(x)$ is the stepwise linear permittivity profile as defined in Fig. 1. For TM polarized light the electromagnetic field is written as $\mathbf{E} = [E_x, 0, E_z]$ and $\mathbf{H} = [0, H_y, 0]$. For 1D problem we look for stationary solutions in the following form:

$$\mathbf{E}(x,z,t) = \mathbf{E}_{\mathrm{NL}}(x)\exp[i(\beta_{\mathrm{NL}}k_0 z - \omega t)], \qquad (1)$$

where $k_0 = 2\pi/\lambda$, λ denotes the free space wavelength and β_{NL} is the effective index of this nonlinear wave. The magnetic field $\mathbf{H}$ has the same form.

First, we find the y-component of the $\mathbf{H}_{\mathrm{NL}}$ field denoted by $H_{\mathrm{NL},y}(x)$ by solving the 1D nonlinear problem along x direction. To initiate the derivation we insert Eq. (1) into Maxwell equations and we use the TM approximation. In our model we introduce the assumption of "transverse nonlinearity" (as used in [6], and it is verified *a posteriori*), that is: $\varepsilon_x = \varepsilon_1 + \alpha E_x^2$ and $\varepsilon_z = \varepsilon_1$. Using this assumption and combining the three TM Maxwell equations, we get:

$$\frac{\partial^2 H_{\mathrm{NL},y}}{\partial x^2} + \varepsilon_1 k_0^2\left(1 - \frac{\beta_{\mathrm{NL}}^2}{\varepsilon_1 + \alpha E_x^2}\right)H_{\mathrm{NL},y} = 0, \qquad (2)$$

All our study is intended to study low power solitons. This implies that the relative permittivity changes due to Kerr effect are small. Therefore, we can write that: $\Delta\varepsilon = \alpha E_x^2 \ll \varepsilon_1$ or equivalently: $\frac{\alpha E_x^2}{\varepsilon_1} \ll 1$. Expanding in Taylor series the nonlinear term, we can write:

$$\frac{\partial^2 H_{\mathrm{NL},y}}{\partial x^2} + \varepsilon_1 k_0^2\left[1 - \frac{\beta_{\mathrm{NL}}^2}{\varepsilon_1}\left(1 - \frac{\alpha E_x^2}{\varepsilon_1}\right)\right]H_{\mathrm{NL},y} = 0. \qquad (3)$$

Using once again the Maxwell equations in the TM case, we obtain an equation that cannot yet be solved due to the dependence of ε_x on E_x^2. However, as stated before the nonlinear permittivity variation are very small ($\Delta\varepsilon < 10^{-3}$) consequently, keeping only main order terms, this equation can be written in form (with the definition $q^2 = \beta_{\mathrm{NL}}^2 - \varepsilon_1$.):

$$\frac{\partial^2 H_{\mathrm{NL},y}}{\partial x^2} - k_0^2 q^2 H_{\mathrm{NL},y} + k_0^2\frac{\alpha\beta_{\mathrm{NL}}^4}{\varepsilon_0^2\varepsilon_1^3 c^2}H_{\mathrm{NL},y}^3 = 0. \qquad (4)$$

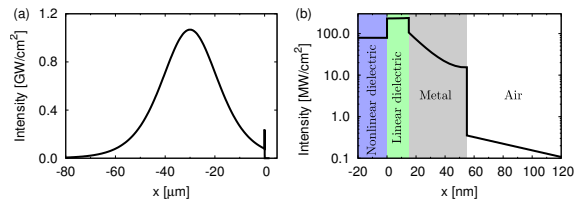

FIGURE 2. Intensity profiles for low power solutions of 1D problem. Full solution (a) and zoom on the plasmon part at the metal interfaces (b). Parameters used are: $\varepsilon_1 = 2.4707^2$, $n_2 = 10^{-13}\mathrm{cm}^2/\mathrm{W}$ (chalcogenide glass), $\varepsilon_2 = 1.443^2$ (silica), $\varepsilon_3 = -96$ (gold), $\varepsilon_4 = 1$ (air), $L = 15$nm, $d = 40$nm, $\lambda = 1.55\mu$m and $x_0 = -30\mu$m.

Finally, using TM Maxwell equations to come back to $E_{\mathrm{NL},x}$, Eq. (4) leads to a nonlinear wave equation valid in the whole structure through the use of the $a(x)$ term:

$$\frac{\mathrm{d}^2 E_{\mathrm{NL},x}}{\mathrm{d}x^2} - k_0^2 q^2(x)E_{\mathrm{NL},x} + k_0^2 a(x)E_{\mathrm{NL},x}^3 = 0, \qquad (5)$$

in which $a(x) = \beta_{\mathrm{NL}}^2\alpha(x)/\varepsilon(x)$ and $q^2(x) = \beta_{\mathrm{NL}}^2 - \varepsilon(x)$. Solution of Eq. (5) provides analytical expressions for $E_{\mathrm{NL},x}(x)$, and Maxwell's equations yield the expressions for the remaining [$E_{\mathrm{NL},z}(x)$ and $H_{\mathrm{NL},y}(x)$] field components. Finally using the field continuity conditions at the interfaces, we get the analytical implicit form of the nonlinear dispersion relation of the 4-layer model:

$$\Phi_+\left(\widetilde{q_4} + \widetilde{q_3}\right)\exp(2k_0\widetilde{q_3}\varepsilon_3 d) + \Phi_-\left(\widetilde{q_4} - \widetilde{q_3}\right) = 0, \qquad (6)$$

$$\Phi_\pm = \left(1 \pm \frac{\widetilde{q_{1\mathrm{NL}}}}{\widetilde{q_3}}\right) + \left(\frac{\widetilde{q_{1\mathrm{NL}}}}{\widetilde{q_2}} \pm \frac{\widetilde{q_2}}{\widetilde{q_3}}\right)\tanh(k_0\widetilde{q_2}\varepsilon_2 L), \qquad (7)$$

where $\widetilde{q_j} = q(x)/\varepsilon(x)$ in the j-th layer, $\widetilde{q_{1\mathrm{NL}}} = \widetilde{q_1}\tanh(k_0\widetilde{q_1}\varepsilon_1 x_0)$ and x_0 denotes the center of the soliton part of the nonlinear wave that appears in the expression for the field in the nonlinear dielectric layer: $E_{\mathrm{NL},x}(x) = [2/a(x)]^{1/2}\widetilde{q_1}\varepsilon_1\,\mathrm{sech}[k_0\widetilde{q_1}\varepsilon_1(x - x_0)]$.

Solving Eq. (6) for β_{NL}, we compute the allowed β_{NL} values of this nonlinear 1D problem that are then used to determine all the field components from Eq. (5). The 1D intensity profile for a peak power around 1 GW/cm^2 is shown in Fig. 2. We verify *a posteriori* that the assumption $|E_z| \ll |E_x|$ is fulfilled in the nonlinear medium (in our example $|E_x|/|E_z| > 50$ for soliton peak intensities lower than 10GW/cm^2).

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We can also study two limiting cases. First we can study the limit of the 4-layer configuration to the 3-layer configuration (which corresponds to $L \to 0$ to get a nonlinear dielectric medium/metal/linear dielectric medium). For $L \to 0$, we have: $\tanh(k_0\widetilde{q_2}\varepsilon_2 L) \to 0$ and Eq.(7) becomes $\Phi_\pm = \left(1 \pm \frac{\widetilde{q_{1\mathrm{NL}}}}{\widetilde{q_3}}\right)$. Using it in Eq. (6), we obtain :

$$\tanh(k_0\widetilde{q_3}\varepsilon_3 d) = -\frac{\widetilde{q_3}(\widetilde{q_{1\mathrm{NL}}} + \widetilde{q_4})}{\widetilde{q_3}^2 + \widetilde{q_{1\mathrm{NL}}}\widetilde{q_4}} \tag{8}$$

which is equivalent to equation (8) in [6].

Secondly, we can study the limit of the 3-layer configuration to a 2-layer configuration (which corresponds to $L = 0$ and $d \to \infty$ to get a nonlinear dielectric medium/metal with $d \to \infty$ /linear dielectric medium). We directly get: $(\widetilde{q_{1\mathrm{NL}}} + \widetilde{q_3})(\widetilde{q_3} + \widetilde{q_4}) = 0$. This condition is fulfilled in two cases. The first one is $\widetilde{q_3} + \widetilde{q_4} = 0$ which, using the definition $\widetilde{q_i} = q_i/\varepsilon_i$, gives the standard dispersion relation for plasmon on metal/linear dielectric interface [1]. The second case gives $\widetilde{q_{1\mathrm{NL}}} + \widetilde{q_3} = 0$ that corresponds to the interface between a nonlinear medium and a metal. Using the definition $\widetilde{q_{1\mathrm{NL}}} = \widetilde{q_1}\tanh(k_0\widetilde{q_1}\varepsilon_1 x_0))$, the equation gives the expression $\frac{q_3}{\varepsilon_3} = -\frac{q_1}{\varepsilon_1}\tanh[k_0 q_1 x_0]$ that is similar to the dispersion relation given by eq. (7) in [14].

TWO EXAMPLES

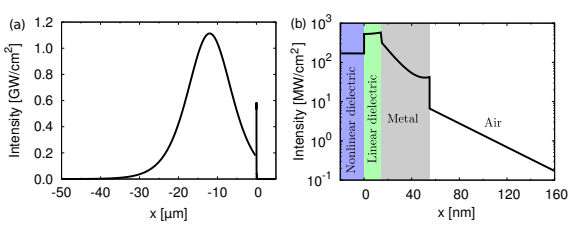

FIGURE 3. Intensity profiles for low power solutions of 1D problem. Full solution (a) and zoom on the plasmon part at the metal interfaces (b). Parameters used are: $\varepsilon_1 = 2.573^2$, $n_2 = 10^{-13}\mathrm{cm}^2/\mathrm{W}$ (chalcogenide glass), $\varepsilon_2 = 1.453^2$ (silica), $\varepsilon_3 = -25$ (gold), $\varepsilon_4 = 1.3^2$ (water), $L = 15$nm, $d = 40$nm, $\lambda = 0.8\mu$m and $x_0 = -12\mu$m.

In the section we briefly describe two examples to illustrate our model. In Fig. 2, the plasmon peak light intensity is one tenth of the soliton one what corresponds to a peak electric field value in the metal layer of approximatively $4.5 \times 10^6\mathrm{V/m}$ which makes it recordable using current near field optics techniques [1]. The propagation losses in the structure can be estimated using the method described in [6, 4]. For the first example shown here (assuming $\Im m(\varepsilon_1) = \Im m(\varepsilon_2) = 10^{-5}$, $\Im m(\varepsilon_3) = 10$), the losses are at the level $\Im m(\beta) \approx 0.8 \times 10^{-5}$, corresponding approximately to 2.8dB/cm.

In Fig. 3, we show an other structure where the external medium is water. It can be seen that the plasmon peak intensity is now around one third of the soliton one. In our talk, for the 4-layer nonlinear model, details will be given on the different solutions and their properties.

CONCLUSION

In conclusion, we have described for the first time a simple 3-layer planar structure made of conventional materials that supports a soliton hybridized by a recordable plasmonic part on one of its tails. These stationary solutions are characterized by low peak powers, comparable to spatial solitons already formed in chalcogenide planar waveguides [13]. To obtain such results, we have developed a vector 1D model for a 4-layer configuration including a nonlinear region.

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REFERENCES

1. S. A. Maier, *Plasmonics: Fundamentals and Applications*, Springer, 2007.
2. Y. S. Kivshar, and G. P. Agrawal, *Optical Solitons, From Fibers to Photonic Crystals*, Academic Press, Amsterdam, 2003.
3. E. Feigenbaum, and M. Orenstein, *Opt. Lett.* **32**, 674 (2007).
4. A. R. Davoyan, I. V. Shadrivov, and Y. S. Kivshar, *Opt. Express* **16**, 21732–21737 (2009).
5. K. Y. Bliokh, Y. P. Bliokh, and A. Ferrando, *Phys. Rev. A* **79**, 41803 (2009).
6. J. Ariyasu, C. T. Seaton, G. I. Stegeman, A. A. Maradudin, and R. F. Wallis, *J. Appl. Phys.* **58**, 2460–2466 (1985).
7. G. I. Stegeman, and C. T. Seaton, *Opt. Lett.* **9**, 235–237 (1984).
8. A. A. Maradudin, "Nonlinear Surface Electromagnetic Waves," in *Optical and Acoustics Waves in Solids – Modern Topics*, edited by M. Borissov, Word Scientific, 1983, pp. 72–142.
9. D. Mihalache, D. Mazilu, and F. Lederer, *Optics Communications* **59**, 391 – 394 (1986), ISSN 0030-4018.
10. A. R. Davoyan, I. V. Shadrivov, and Y. S. Kivshar, *Opt. Express* **16**, 21209 (2008).
11. I. D. Rukhlenko, A. Pannipipitiya, and M. Premaratne, *Opt. Lett.* **36**, 3374 (2011).
12. A. Zakery, and S. R. Elliott, *Optical Nonlinearities in Chalcogenide Glasses*, Springer, 2007.
13. M. Chauvet, G. Fanjoux, K. P. Huy, V. Nazabal, F. Charpentier, T. Billeton, G. Boudebs, M. Cathelinaud, and S.-P. Gorza, *Opt. Lett.* **34**, 1804 (2009).
14. V. M. Agranovich, V. S. Babichenko, and V. Y. Chernyak, *Sov. Phys. JETP* **32**, 512–515 (1980).

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